التكنولوجيا التطبيقية
APPLIED TECHNOLOGY
وزارة التربية والتعليم والتعليم الفني

Field Training Guide
Unit : 9
AI Principles and Applications

Prepared by
IT General

Academic Year
2025 – 2026

All rights reserved to ATS-MU

Unit	**9**	
Name	**AI Principles and Applications**	
Goals / Outcomes	By the end of this unit, students will be able to: ➢ Understand that AI is a pattern-matching prediction system, not a thinking human mind ➢ Build a functional multimodal Smart Assistant using PictoBlox ➢ Analyze why AI systems make mistakes and how to diagnose failures ➢ Evaluate AI outputs critically, considering confidence scores and error types ➢ Apply ethical reasoning to AI applications, recognizing bias and security risks ➢ Explain AI concepts to others using accurate terminology	
Knowledge	**Code**	**Description**
	TK-A1	**1- Explain AI as a data-driven system including predictions, confidence, and failures.**
	TK-A2	**2- Design AI applications and their evaluation methods.**
Skill	**Code**	**Description**
	TC-A1	**Build AI perception applications with correct configuration and threshold adjustment.**
	TC-A2	**Analyze AI performance including failures, errors, and data impact.**

Copy rights reserved to ATS-MU

LESSON 1 PRACTICAL TASK	"The Confidence Detective"
Learning Skills Covered:	**By the end of this task, students will be able to:** ➢ Training an AI model with data ➢ Understanding confidence scores ➢ Observing how more data improves predictions ➢ Testing model stability

Task Overview

You will train an Image Classifier in PictoBlox with very few images, then add more images to see how confidence improves. This shows how AI learns from data.

Materials Needed

- PictoBlox installed on computer
- Two simple objects (e.g., a bottle and a book, or a pen and a phone)
- Camera (built-in or external)
- Activity worksheet

Steps :

Step 1: Create Your Project

1. **Open PictoBlox**
2. **Click "Create New Project"**
3. **Go to Extensions** (bottom left)
4. **Add "AI Image Classifier"** extension

Step 2: Create Your Classes

1. **Click on "Machine Learning/Classifier" panel**
2. **Create Class 1:** Type "Bottle" (or your first object)
3. **Create Class 2:** Type "Book" (or your second object)

Step 3: Add MINIMAL Training Data (The Weak Model)

Important: We want the model to be WEAK first!

Copy rights reserved to ATS-MU

1. **For Class 1 (Bottle):**
    - Take ONLY 1 or 2 photos
    - Keep the background simple
    - Take photos from the same angle
2. **For Class 2 (Book):**
    - Take ONLY 1 or 2 photos
    - Keep the background simple
    - Take photos from the same angle
3. **Click "Train Model"** button
    - Wait for training to finish (should be quick)

Step 4: Test Your Weak Model

1. **Find a NEW image** of a bottle (not used in training)
    - You can take a new photo or find one online
2. **Test the model:**
    - Click "Test" or upload your new image
    - Record the result

Test Image	Predicted Label	Confidence Score
New Bottle		
New Book		

3. **Test again** with a slightly changed condition:
    - Move the object slightly
    - Change lighting a little
    - Rotate the object

Step 5: Add MORE Training Data (The Strong Model)

1. **For Class 1 (Bottle):**
    - Add 5-8 MORE photos
    - Use DIFFERENT conditions:

Copy rights reserved to ATS-MU

        - Different lighting (bright, dim)
        - Different angles (front, side, top)
        - Different backgrounds (table, floor, wall)
    2. **For Class 2 (Book):**
        - Add 5-8 MORE photos
        - Use the same variations
    3. **Click "Train Model" again**
        - Wait for training to finish (takes a little longer now)

Step 6: Test Your Strong Model

1. **Use the SAME test images** from Step 4

Step 7: The Data Drift Experiment

1. **Take the bottle to a DIFFERENT ROOM** or place
2. **Take a photo** of the same bottle in the new environment
3. **Test the model** with this new photo

Step 8: Add Decision Logic (Optional Challenge)

Now let's make the AI say different things based on confidence.

1. **Go to "Variables"** and create a variable called "Confidence"
2. **Build this script:**

text

```
When Green Flag Clicked
  Analyze object from stage
  Set Confidence to confidence value
  If Confidence > 80 then
    Say "I am sure this is a " + predicted label
  Else if Confidence > 50 then
    Say "I think this is a " + predicted label
  Else
    Say "I'm not sure what this is"
```

© Copy rights reserved to ATS-MU

3. **Test with different images:**
    - Clear image (should say "I am sure")
    - Blurry or dark image (should say "I think")
    - Very confusing image (should say "I'm not sure")

LESSON 2 PRACTICAL TASK	**"The Vision Stress Tester"**
Learning Skills Covered:	**By the end of this task, students will be able to:** ➢ Testing object detection under different conditions ➢ Observing how lighting, angle, and occlusion affect predictions ➢ Understanding bounding boxes ➢ Diagnosing pipeline failures

Task Overview

You will use PictoBlox Object Detection to test how environmental changes affect AI vision. You'll create a "Vision Stress Report" documenting what causes the AI to fail.

Materials Needed

- PictoBlox installed on computer
- 3-4 simple objects (e.g., bottle, book, phone, toy)
- Camera (built-in or external)
- Activity worksheet

Step 1: Set Up Object Detection

1. **Open PictoBlox**
2. **Go to Extensions**
3. **Add "Object Detection"** extension
4. **Click "Open Recognition Window"**
5. **Turn on camera** (or enable image upload)

Step 2: Build Your Detection Script

Create this simple script:

Copy rights reserved to ATS-MU

When Green Flag Clicked

Forever

Recognize image features in image from stage

Show bounding box

Say "I see: " + recognized object 1 name

Say "Confidence: " + recognized object 1 confidence

Wait 1 second

Make sure: Bounding boxes appear around detected objects

Step 3: Baseline Test (Good Conditions)

1. **Place an object** in good lighting, clear view, clean background
2. **Run your script**
3. **Record the results:**

Step 4: The Lighting Stress Test

Test the SAME object under different lighting conditions:

Step 5: The Angle Stress Test

Test the SAME object at different angles:

Step 6: The Occlusion Stress Test (Hidden Parts)

Test the SAME object with parts hidden:

Step 7: The Background Stress Test

Test the SAME object with different backgrounds:

Step 8: The Motion Stress Test

Test the SAME object while moving:

Step 9: Classification vs. Detection Challenge

Now, test a scene with MULTIPLE objects:

© Copy rights reserved to ATS-MU

LESSON 3 PRACTICAL TASK	"The Face Security System"
Learning Skills Covered:	**By the end of this task, students will be able to:** ➢ Face detection vs. face recognition ➢ Enrolling a user ➢ Understanding similarity scores ➢ Adjusting thresholds ➢ Testing spoofing attacks

Task Overview

You will build a face recognition security system in PictoBlox. You'll enroll yourself as the "authorized user," test different thresholds, and try to trick the system with photos.

Materials Needed

- PictoBlox installed on computer
- Camera (built-in or external)
- Printed photo of yourself (or photo on phone screen)
- Activity worksheet

Step 1: Set Up Face Recognition

1. **Open PictoBlox**
2. **Go to Extensions**
3. **Add "Face Detection" or "Face Recognition"** extension
4. **Turn on camera**

Step 2: Enroll Yourself (Create Your Face Print)

Important: This is the "Garbage In, Garbage Out" moment! Make it good!

Golden Rules for Enrollment:

- Good lighting (not too dark, not too bright)
- Front-facing (looking straight at camera)
- Neutral expression (not extreme smile/frown)
- Steady camera (no motion blur)

1. **Click "Enroll" or "Add Face"**

© Copy rights reserved to ATS-MU

2. **Capture 15-20 images** of your face
3. **Label it** as "Authorized User" or your name
4. **Train the model**

Step 3: Build Your Security System

Create this script:

```
When Green Flag Clicked
  Create variable "Threshold"
  Set Threshold to 80
    Forever
    Recognize face from camera
    If similarity score > Threshold then
      Say "Access Granted: Welcome!"
      Change sprite color to green
    Else
      Say "Access Denied: Not recognized"
      Change sprite color to red
    Wait 2 seconds
```

Step 4: Test Your System (Baseline)

Calculate your average similarity score: _____%

Step 5: Test Different Thresholds

Change the Threshold variable and test:

Step 6: Test Environmental Conditions

Test YOUR face under different conditions:

Step 7: The Spoofing Test (Try to Trick the System!)

WARNING: This shows how face recognition can be vulnerable!

1. **Print a photo** of the authorized user (or display on another phone screen)
2. **Hold the photo up to the camera**
3. **Test the system**

© Copy rights reserved to ATS-MU

Step 8: Add Liveness Detection (Anti-Spoofing)

Now let's make it harder to trick!

Modify your script to require a blink:

```text
When Green Flag Clicked
  Create variable "Threshold"
  Set Threshold to 80
    Forever
    Recognize face from camera
    If similarity score > Threshold then
      Say "Face recognized! Please blink to confirm"
      Wait for blink (check eyelid landmarks)
      If blink detected then
        Say "Access Granted"
        Change sprite color to green
      Else
        Say "Liveness check failed"
    Else
      Say "Access Denied: Not recognized"
      Change sprite color to red
    Wait 2 seconds
```

Test again:

Step 9: The Bias Simulation (Optional)

1. **Enroll TWO people:**
   - Person A: 25 training images
   - Person B: 5 training images
2. **Test both under same conditions:**

Was there a difference? Which person was more likely to be rejected?

Copy rights reserved to ATS-MU

LESSON 4 PRACTICAL TASK	"The Gesture Controller"
Learning Skills Covered:	**By the end of this task, students will be able to:** ➢ Pose estimation (tracking body joints) ➢ Creating simple gesture rules ➢ Understanding temporal stability ➢ Testing real-time performance

Task Overview

You will build a gesture-controlled Smart Assistant in PictoBlox. You'll create a "raise hand" gesture that triggers an action, and test how well it works under different conditions.

Materials Needed

- PictoBlox installed on computer
- Camera (built-in or external)
- Space to move (standing in front of camera)
- Activity worksheet

Step 1: Set Up Pose Detection

1. **Open PictoBlox**
2. **Go to Extensions**
3. **Add "Human Body Detection" or "Pose Detection"** extension
4. **Turn on camera**
5. **Stand in front of camera** so your whole body is visible

Step 2: Visualize Your Skeleton

1. **Add a script to show keypoints:**

When Green Flag Clicked

Forever

Detect human pose from camera

Show skeleton

Show keypoints

© Copy rights reserved to ATS-MU

Wait 0.1 seconds

2. **Stand in front of camera and observe:**
   - Can you see dots on your joints (shoulders, elbows, wrists, hips, knees, ankles)?
   - Are they connected by lines?

Step 3: Find Your Joint Coordinates

Let's track one joint and see its coordinates change as you move.

Create a script to track your wrist:

text

```
When Green Flag Clicked
  Forever
    Detect human pose from camera
    Set "Wrist X" to x position of right wrist
    Set "Wrist Y" to y position of right wrist
    Say "Wrist position: (X: " + Wrist X + " Y: " + Wrist Y + ")"
    Wait 0.5 seconds
```

Move your arm and watch the numbers change:

Step 4: Create a Simple Gesture Rule

Let's make a "Raise Right Hand" detector.

The rule:

If right wrist is ABOVE right shoulder → hand is raised

Create this script:

text

```
When Green Flag Clicked
  Create variable "HandRaised"
  Set HandRaised to 0
    Forever
    Detect human pose from camera
```

Copy rights reserved to ATS-MU

```
If (y position of right wrist) < (y position of right shoulder) then
  Set HandRaised to 1
  Say "Hand is RAISED!"
  Change sprite color to green
Else
  Set HandRaised to 0
  Say "Hand is down"
  Change sprite color to red
Wait 0.1 seconds
```

Test it:

- Lower your arm → should say "Hand is down" (red)
- Raise your arm → should say "Hand is RAISED!" (green)

Step 5: Add Temporal Stability (Prevent False Triggers)

Problem: The gesture might trigger for just one frame (jitter) when you didn't actually mean to raise your hand.

Solution: Require the gesture to be held for several frames.

Create this improved script:

text

```
When Green Flag Clicked
  Create variable "HandRaisedCount"
  Set HandRaisedCount to 0
   Forever
    Detect human pose from camera

    If (y position of right wrist) < (y position of right shoulder) then
      Set HandRaisedCount to HandRaisedCount + 1
    Else
```

Copy rights reserved to ATS-MU

```
    Set HandRaisedCount to 0
      If HandRaisedCount > 10 then
    Say "GESTURE DETECTED! Hand raised!"
    Change sprite color to green
    Wait 2 seconds
  Else
    Say "Waiting for gesture..."
    Change sprite color to red
  Wait 0.1 seconds
```

Test it:

- Quickly raise and lower your hand → should NOT trigger
- Raise and HOLD for 1-2 seconds → should trigger

Step 6: Create a Second Gesture

Let's add "Both Hands Up" as a second gesture.

The rule:

If both wrists are above their shoulders → emergency mode

Add to your script:

text

```
If (y of right wrist) < (y of right shoulder) AND
  (y of left wrist) < (y of left shoulder) then
  Say "EMERGENCY MODE ACTIVATED!"
  Change sprite color to blue
  Play sound "alert" (if available)
```

Test both gestures:

Step 7: Test Environmental Conditions

Test your gesture detector under different conditions:

Which condition caused the most problems?

© Copy rights reserved to ATS-MU

Step 8: Test Real-Time Performance

Test 1: Speed Test

- Raise your hand as fast as you can
- How long does it take for the system to respond?

Test 2: Load Test

1. While pose detection is running, add a SECOND sprite that moves around constantly
2. Add MORE sprites (5-10) moving around
3. Test your gesture again

Step 9: Connect to Your Smart Assistant

Now let's make your gesture control something!

Create this final script:

text

```
When Green Flag Clicked
  Create variable "HandRaisedCount"
  Set HandRaisedCount to 0
    Forever
    Detect human pose from camera
        If (y of right wrist) < (y of right shoulder) then
      Set HandRaisedCount to HandRaisedCount + 1
    Else
      Set HandRaisedCount to 0
        If HandRaisedCount > 10 then
      Say "GESTURE DETECTED!"
      Change sprite color to green
      // Add your action here!
      // Example: broadcast a message to turn on lights
      Broadcast "Activate"
```

© Copy rights reserved to ATS-MU

Wait 2 seconds

Else

Say "Raise your right hand to activate"

Wait 0.1 seconds

Your action could be:

- Change background color
- Play a sound
- Show a message
- Control another sprite
- (If you have other modules) turn on a virtual light

LESSON 5 PRACTICAL TASK	**"The Voice Assistant"**
Learning Skills Covered:	**By the end of this task, students will be able to:** ➢ Speech recognition ➢ Natural language understanding ➢ Handling paraphrases ➢ Detecting hallucination ➢ Safe response design

Task Overview

You will build a voice-controlled Smart Assistant in PictoBlox that responds to voice commands. You'll test how it handles different phrasings and learn why it sometimes gives wrong answers.

Materials Needed

- PictoBlox installed on computer
- Microphone (built-in or external)
- Quiet room for testing
- Activity worksheet

Step 1: Set Up Speech Recognition

1. **Open PictoBlox**
2. **Go to Extensions**

Copy rights reserved to ATS-MU

3. **Add "Speech to Text"** extension
4. **Check your microphone is working**

Step 2: Build a Simple Voice Assistant

Create this script:

text

```
When Green Flag Clicked
  Say "Say a command"
    Forever
    Recognize speech for 5 seconds
    Set "UserCommand" to speech recognition result
    Say "You said: " + UserCommand
    Wait 2 seconds
```

Test it:

1. Click the green flag
2. Say something clearly (e.g., "Hello")
3. See what the AI thinks you said

Step 3: Create Simple Command Responses

Let's make your assistant respond to specific commands.

Create this script:

text

```
When Green Flag Clicked
  Say "Say a command: lights, weather, or greeting"
  Forever
    Recognize speech for 5 seconds
    Set "UserCommand" to speech recognition result

    If UserCommand contains "light" then
      Say "Turning lights on!"
```

© Copy rights reserved to ATS-MU

```
    Change backdrop brightness to 100
  Else if UserCommand contains "weather" then
    Say "It's sunny today!"
  Else if UserCommand contains "hello" or "hi" then
    Say "Hello! Nice to meet you!"
  Else
    Say "I don't understand that command"
  Wait 2 seconds
```

Test these commands:

Step 4: Test Paraphrases (Different Ways to Say Same Thing)

Test different ways to say the same thing:

Step 5: Test Under Different Conditions

Test 1: Quiet vs. Noisy

Test 2: Different Speech Styles

Step 6: Add Confidence Display

Let's see how confident the AI is about what it heard.

Modify your script:

text

```
When Green Flag Clicked
  Say "Say a command"
    Forever
    Recognize speech for 5 seconds
    Set "UserCommand" to speech recognition result
    Set "Confidence" to confidence value of speech result

    Say "You said: " + UserCommand
    Say "I am " + Confidence + "% sure"
```

© Copy rights reserved to ATS-MU

```
    If Confidence > 80 then
      // Process command
    Else
      Say "I'm not sure what you said. Please repeat."
    Wait 2 seconds
```

Test with clear and unclear speech:

Step 7: Create a Safe Assistant (With Confirmation)

Add confirmation for important actions:

text

```
When Green Flag Clicked
  Say "Say a command"
  Forever
    Recognize speech for 5 seconds
    Set "UserCommand" to speech recognition result
    If UserCommand contains "light" then
      Say "Did you say to turn on the lights? Say YES or NO"
      Recognize speech for 3 seconds
      If speech result contains "yes" then
        Say "Turning lights on!"
      Else
        Say "Command cancelled"
    Else
      Say "Command not recognized"
    Wait 2 seconds
```

Test it:

- Say "Turn on lights"
- When asked to confirm, say "yes" or "no"

Why is confirmation useful?

© Copy rights reserved to ATS-MU